Online Voting System – MIS Application

Domain:

Election / Voting Management System

Problem Statement:

Traditional voting systems in small organizations, colleges, or clubs are usually conducted manually. Manual voting can lead to several problems such as inaccurate vote counting, duplicate voting, and difficulty in maintaining records. It also lacks transparency and efficiency.

To overcome these problems, an **Online Voting System** is developed using Python. The system allows registered users to vote securely through a simple interface. It maintains a record of users, candidates, and votes using text files. The system ensures that only authenticated users can vote and prevents multiple votes from the same user.

This MIS application helps manage voting data efficiently and provides real-time results.

Objectives:

The objectives of the Online Voting System are:

- To develop a **simple voting management system using Python**.
- To maintain voter and candidate data using **text files**.
- To allow **user registration and login authentication**.
- To ensure that **each user can vote only once**.
- To maintain **candidate vote counts automatically**.
- To display **election results clearly**.

Dataset:

The system uses **three text files as a database**.

1. Users Dataset (users.txt)

Stores registered user credentials.

Example:

Salman Khan,katrina69
Vijay,trisha11
Ranbir Kapoor,deepika99
Anjali,rohan6969
Narendra Modi,giorgia00
Karan Johar,ranveer96
Karthik Aryan,pedo16
Sharadha Kapoor,Aditya7
Kajol Devgan,SRK67
Nayantara,STR99

Fields:

- Username
- Password

2. Candidates Dataset (candidates.txt)

Stores candidate names and vote counts.

Example:

Osama Bin Laden,4
Mahatma Gandhi,2
Rehman Dakait,3
Charlie Kirk,1

Fields:

- Candidate Name
- Vote Count

3. Votes Dataset (votes.txt)

Stores voting records.

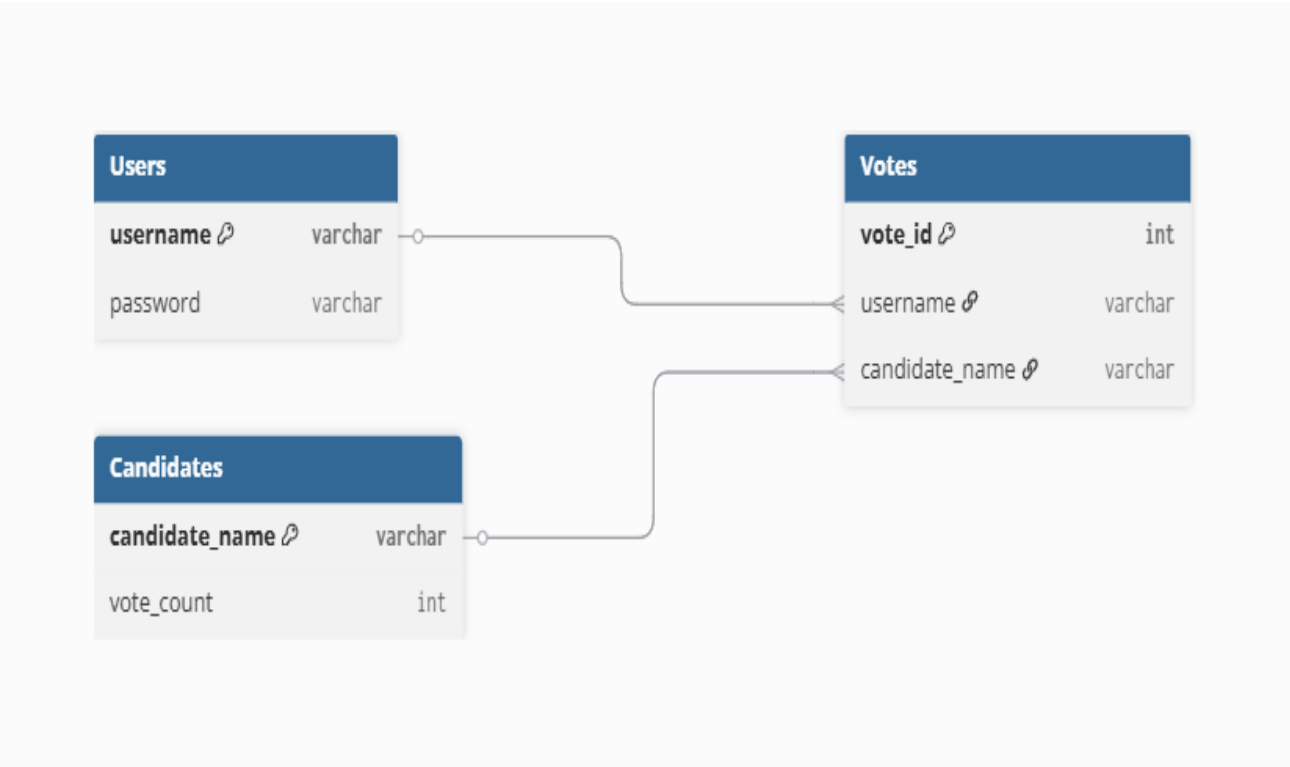
Example:

Salman Khan,Osama Bin Laden
Vijay,Mahatma Gandhi
Ranbir Kapoor,Rehman Dakait
Anjali,Rehman Dakait
Narendra Modi,Mahatma Gandhi
Karan Johar,Charlie Kirk
Karthik Aryan,Osama Bin Laden
Sharadha Kapoor,Osama Bin Laden
Kajol Devgan,Rehman Dakait
Nayantara,Osama Bin Laden

Fields:

- Username
- Candidate voted for

ER Diagram:



Code:

copy_of_welcome_to_colab.py

C: > Users > Dharshan > C:\Users\Dharshan\Downloads\cia\copy_of_welcome_to_colab.py

```python
# File paths
USER_FILE = "users.txt"
CANDIDATE_FILE = "candidates.txt"
VOTES_FILE = "votes.txt"


# Initialize files
def initialize_files():
    files = [USER_FILE, CANDIDATE_FILE, VOTES_FILE]
    for file in files:
        with open(file, 'a'):
            pass


# Register new voter
def register_voter():
    username = input("Enter username: ")
    password = input("Enter password: ")

    with open(USER_FILE, 'r') as f:
        users = [line.strip().split(',') for line in f.readlines()]

    for user in users:
        if user[0] == username:
            print("Username already exists!")
            return

    with open(USER_FILE, 'a') as f:
        f.write(f"{username},{password}\n")

    print("Registration successful!")


# Authenticate user
def authenticate():
    username = input("Enter username: ")
    password = input("Enter password: ")

    with open(USER_FILE, 'r') as f:
```

copy_of_welcome_to_colab.py

C: > Users > Dharshan > Downloads > cia > copy_of_welcome_to_colab.py > ...

```python
def authenticate():
        users = [line.strip().split(',') for line in f.readlines()]

    for user in users:
        if user[0] == username and user[1] == password:
            print("Login successful!")
            return username

    print("Invalid credentials!")
    return None


# Add candidate
def add_candidates():
    name = input("Enter candidate name: ")

    with open(CANDIDATE_FILE, 'r') as f:
        candidates = [line.strip().split(',')[0] for line in f.readlines()]

    if name in candidates:
        print("Candidate already exists!")
        return

    with open(CANDIDATE_FILE, 'a') as f:
        f.write(f"{name},0\n")

    print("Candidate added successfully!")


# Cast vote
def cast_vote(voter):

    with open(VOTES_FILE, 'r') as vf:
        votes = [line.strip().split(',') for line in vf.readlines()]

    for v in votes:
        if v[0] == voter:
            print("You have already voted!")
            return
```

copy_of_welcome_to_colab.py

C: > Users > Dharshan > Downloads > cia > copy_of_welcome_to_colab.py > ...

```python
def cast_vote(voter):
    with open(CANDIDATE_FILE, 'r') as cf:
        candidates = [line.strip().split(',') for line in cf.readlines()]

    print("Candidates:")
    for c in candidates:
        print(c[0])

    choice = input("Enter candidate name to vote: ")

    updated_candidates = []
    found = False

    for c in candidates:
        name, vote_count = c
        if name == choice:
            vote_count = str(int(vote_count) + 1)
            found = True

        updated_candidates.append(f"{name},{vote_count}\n")

    if not found:
        print("Invalid candidate!")
        return

    with open(CANDIDATE_FILE, 'w') as cf:
        cf.writelines(updated_candidates)

    with open(VOTES_FILE, 'a') as vf:
        vf.write(f"{voter},{choice}\n")

    print("Vote cast successfully!")


# Display results
def show_results():
    with open(CANDIDATE_FILE, 'r') as cf:
        candidates = [line.strip().split(',') for line in cf.readlines()]

```

copy_of_welcome_to_colab.py

C: > Users > Dharshan > Downloads > cia > copy_of_welcome_to_colab.py > ...

```python
def show_results():
    print("\nElection Results:")
    for name, votes in candidates:
        print(f"{name}: {votes} votes")


# Main menu
def main():

    initialize_files()

    while True:

        print("\n1. Register")
        print("2. Login & Vote")
        print("3. Add Candidate")
        print("4. View Results")
        print("5. Exit")

        choice = input("Enter choice: ")

        if choice == "1":
            register_voter()

        elif choice == "2":
            voter = authenticate()
            if voter:
                cast_vote(voter)

        elif choice == "3":
            add_candidates()

        elif choice == "4":
            show_results()

        elif choice == "5":
            print("Exiting program.")
            break

```

copy_of_welcome_to_colab.py

C: > Users > Dharshan > Downloads > cia > copy_of_welcome_to_colab.py > ...

```python
def main():
        else:
            print("Invalid choice. Try again.")


if __name__ == "__main__":
    main()
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Dharshan> & C:/Users/Dharshan/AppData/Local/Programs/Python/Python314/python.exe c:/Users/Dharshan/Downloads/cia/copy_of_welcome_to_colab.py

1. Register
2. Login & Vote
3. Add Candidate
4. View Results
5. Exit
Enter choice: 3
Enter candidate name: Modi
Candidate added successfully!

1. Register
2. Login & Vote
3. Add Candidate
4. View Results
5. Exit
Enter choice: 1
Enter username: adi
Enter password: 123
Registration successful!

1. Register
2. Login & Vote
3. Add Candidate
4. View Results
5. Exit
Enter choice: 2
Enter username: adi
Enter password: 123
Login successful!
Candidates:
Modi
Enter candidate name to vote: Modi
Vote cast successfully!

1. Register
2. Login & Vote
3. Add Candidate
4. View Results
5. Exit
Enter choice: 4
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Election Results:
Enter choice: 4

Election Results:
Election Results:
Modi: 1 votes
```

Conclusion:

The Online Voting System developed in this project demonstrates how a simple Management Information System (MIS) can be used to manage and process voting activities efficiently. The system allows users to register, log in securely, and cast their vote for a candidate. It also ensures that each user can vote only once, which helps prevent duplicate voting.

The application uses text files to store user data, candidate information, and voting records. This makes the system simple and easy to implement without requiring complex databases. The system also automatically updates vote counts and displays election results, improving accuracy and transparency.

Overall, the project shows how basic programming concepts such as file handling, user authentication, and data management can be applied to build a practical voting management system.